

Medical Visualizations with Dynamic Shape and Depth Cues

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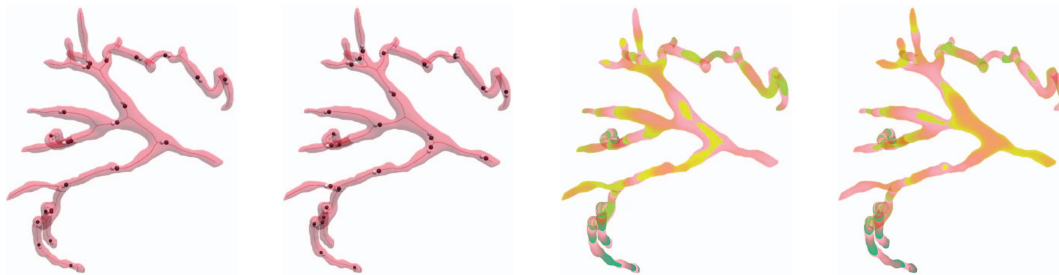


Figure 1: The use of dynamic visualization techniques such as *Object Flow* (a,b) and *Wave Propagation* (c,d), here applied to a model of a blood vessel, can support AR users by conveying shape and depth information through motion cues.

ABSTRACT

Navigated surgery enables physicians to perform complex tasks assisted by virtual representations of surgical tools and anatomical structures visualized using intraoperative medical images. Integrating Augmented Reality (AR) in these scenarios enriches the virtual information presented to the surgeon by utilizing a wide range of visualization techniques. In this work, we introduce a novel approach to conveying additional depth and shape information of the augmented content using dynamic visualization techniques. Compared to existing works, these techniques allow users to gather information not only from pictorial but also from kinetic depth cues.

Index Terms: Human-centered computing—Visualization—Visualization techniques—; Human-centered computing—Visualization—Empirical studies in visualization—

1 INTRODUCTION

Over the past several decades, many surgical procedures have shifted from traditional open approaches to minimally-invasive surgery (MIS), which has several benefits for the patient. However, the advantages for the patient are accompanied by perceptual problems for the surgeons, as they lose the direct view of the surgical site and have no possibility for haptic verification [3]. This lack of direct view affects the depth and shape perception of relevant structures, as some visual cues are unavailable. Augmented Reality (AR) has proven valuable in assisting physicians during MIS procedures, e.g., combining laparoscopic images with virtual content. By including preoperative information, such as patient-specific medical images

or surgical planning, AR can effectively introduce new sources of information to compensate for lost data in MIS compared to open approaches [1]. By tracking the position of the laparoscope, the surgical instruments, and the patient, it is possible to combine the laparoscopic camera image with virtual content in a spatially and temporally consistent way. However, integrating virtual content at the correct position over time in the video stream is insufficient to provide reliable information to the users of AR during MIS. In addition, it is necessary that the users correctly understand the position, shape, and depth of the virtual objects. Therefore, the selection of visualization techniques to present the virtual content plays a determinant role in the system's efficacy.

The human visual system estimates depth using various visual cues. AR systems can replicate some of them, but conflicting information results in the incorrect perception of the object's depth. Existing approaches, such as *illustrative visualization techniques*, often use non-photorealistic methods to present virtual content [2]. Although these techniques can provide additional information during MIS, they consistently rely on static perceptual cues. This work discusses whether the integration of motion cues impacts shape and depth perception in medical AR applications.

2 DYNAMIC VISUALIZATION METHODS

We introduce two novel visualization techniques that use motion as an additional cue to convey shape and depth.

We define **Object Flow** as a dynamic illustrative visualization technique that propagates auxiliary objects through the main object. They move along a path in the middle of the main object, which is also displayed to improve the shape perception further. As we use flowing objects with an asymmetric silhouette, each of them indicates a direction and distance over time. In our implementation, we additionally render the surface semi-transparently with a fixed transparency value. Fig. 1a and 1b illustrate this technique using two sequential images indicating the dynamic behavior.

Our **Wave Propagation** approach is a dynamic illustrative visualization technique that animates view-dependent feature lines (grid lines) combined with Chroma Depth (Viridis colormap). Chroma

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Table 1: Mean and standard deviation for the objective and subjective results collected from the shape and depth perception studies.

	Shape Accuracy	Shape Time	Shape SEQ	Shape Favorite	Shape Least Favorite	Depth Accuracy	Depth SUS	Depth TLX	Depth Favorite	Depth Least Favorite
Fresnel	0.64 ± 0.21	34.52 ± 24.11	4.29 ± 0.86	0.08	0.05	0.44 ± 0.20	49.21 ± 16.98	3.22 ± 1.10	0.05	0.13
Projection	0.59 ± 0.22	42.69 ± 30.89	5.17 ± 0.96	0.04	0.41	0.53 ± 0.25	46.05 ± 19.17	3.52 ± 1.16	0.00	0.73
Hatching	0.69 ± 0.26	34.80 ± 25.16	4.15 ± 1.37	0.28	0.09	0.60 ± 0.26	55.79 ± 18.43	2.89 ± 1.14	0.26	0.07
Object Flow	0.62 ± 0.25	40.91 ± 30.13	3.81 ± 1.29	0.48	0.05	0.46 ± 0.32	56.18 ± 19.16	2.83 ± 1.08	0.42	0.00
Wave	0.63 ± 0.28	44.58 ± 44.50	4.89 ± 1.52	0.12	0.41	0.59 ± 0.24	54.08 ± 12.94	3.20 ± 1.05	0.26	0.07

Depth provides explicit pictorial depth cues, while the feature lines support shape perception. By moving the lines along the view direction at a constant speed, every part of the surface will eventually get emphasized. To further indicate the shape, we added a trail with decreasing intensity resembling a wave propagating along the surface of the virtual objects (with the grid line indicating the maximum). As for *Object Flow*, we applied semi-transparent surface rendering with a fixed transparency to the model, as depicted in Fig. 1c and 1d.

3 USER STUDY

We compared the proposed techniques to state-of-the-art visualization techniques in a two-part user study focusing on shape and depth perception. We compared: (i) Fresnel-Edge-Highlighting, (ii) Hatching with Chroma Depth (Viridis colormap), (iii) Projection on Auxiliary Planes, (iv) Wave Propagation, and (v) Object Flow.

In the **Shape Perception** study, participants identified corresponding 3D models of similar blood vessels. This portion of the study was administered as an online application with 24 participants ($M_{age} = 28.1, SD_{age} = 4.8$). After a training session, participants were asked to match a reference vessel against a set of five similar structures represented using one of the five visualization techniques. This task was repeated five times per visualization technique and with five model variations. We used the accuracy of the participants' answers and time to completion as objective measures. In addition, each task was assessed using a single ease question (SEQ) on a 7-point Likert scale (1-“very easy” to 7-“very difficult”).

The **Depth Perception** study was administered as an on-site study in the hospital. After a graphical explanation of the task, 19 participants ($M_{age} = 29.6, SD_{age} = 6.6$) watched a video sequence from a laparoscopic rectum resection recorded with a digital video recorder and an externally tracked laparoscope. The participants were presented with a 3D model of a blood vessel superimposed on the laparoscopic video. At different moments of the video, we introduced a pair of points located at different depths in the vessel tree and asked the participants to determine the closest point to the viewer. A total of five point-pairs for each visualization technique were introduced during the sequence. After this task, participants completed subjective questionnaires for every visualization technique. We collected the accuracy of the estimation task for each answer as objective measures and used the NASA-TLX and System Usability Scale (SUS) questionnaires as subjective measures.

4 RESULTS

We report the means and standard deviations of the accuracy (ratio between correct answers and number of tasks), time (in seconds), and subjective measures for each visualization technique. For the NASA-TLX, we tested the raw-scores (as the average of all sub-scales) and the individual sub-scales. We also investigated statistical significance using Friedman tests with $p < 0.05$ and calculated Bonferroni-corrected pairwise comparisons for significant differences. We replaced in total 20 outliers in the time measurements that were greater than three standard deviations of each condition.

Shape Perception. The Friedman tests did not reveal statistically significant differences for accuracy ($p = 0.65$) and time ($p = 0.075$) for this portion of the study (see Table 1). However, a Friedman

test showed a statistically significant difference in the SEQ between the visualizations ($p \ll 0.01$). A Bonferroni-corrected comparison revealed a statistically significant difference between *Projection* (harder) and *Object Flow* (easier) ($p = 0.039$).

Depth Perception. The Friedman test did not reveal statistical significant differences for the accuracy ($p = 0.28$) in this study (see Table 1). However, a Friedman test showed a difference in the SUS values ($p = 0.033$). The Bonferroni-corrected comparison revealed a difference of statistical significance between *Projection* (lower usability) and *Hatching* (higher usability) ($p = 0.044$). The TLX scores were of non-significant difference ($p > 0.05$) except in the effort category ($p = 0.0077$). According to the Bonferroni-corrected comparisons, there was a significant difference between *Projection* (more effort) and *Object Flow* (less effort) ($p = 0.0095$).

5 DISCUSSION

In both studies, *Wave Propagation* and *Hatching with Chroma Depth* performed well. Both techniques indicate depth with color.

For the *Wave Propagation* and *Object Flow* techniques, the participants watched an animation, which, we believe, increased the time required to complete the task. In contrast, the complexity of the *Projection* method, as indicated by the high TLX score and the participants' free text comments, may explain why participants spent a lot of time on it. Lastly, the *Fresnel* method contained less visual information. Therefore, taking longer would not have helped.

In both studies, the participants indicated that *Object Flow* was their favorite (see Table 1), with the lowest reported task load and highest usability and ease of use.

6 CONCLUSION

The choice of visualization technique in medical AR is complex. There are multiple surgical requirements, but also physicians' preferences and performance differences. While we measured differences in objective and subjective measures, the visualization techniques performed similarly, especially in the accuracy measurement, despite their stark visual differences. Dynamic visualization techniques provide additional motion cues, but their correct interpretation by the viewer may be demanding. More research is needed to fully understand the impact of medical AR visualization techniques, e.g., the long-term effects of using a specific visualization technique or the influence of task complexity and physician experience.

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